AGILE DEVELOPMENT PROCESS AND DEVOPS

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COURSE CODE: ISWE406L

SLOT: C1+TC1

APPOINTMENT BOOKING SYSTEM USING CI/CD PIPELINE

1. Introduction:

Appointment booking systems are widely used in healthcare and service industries to manage scheduling efficiently. Ensuring that appointments are booked correctly without conflicts is critical to avoid confusion and improve service quality.

This project focuses on developing an Appointment Booking System with automated testing and deployment using a CI/CD pipeline.

2. Problem Statement:

To ensure that appointment booking logic works correctly and prevents double booking and invalid time slots before releasing updates.

3. Objectives:

- To develop appointment scheduling logic using Java
- To prevent double booking of time slots
- To validate valid and invalid appointment timings
- To implement automated testing using JUnit
- To automate build and deployment using CI/CD pipeline
- To deploy the system using Docker and Kubernetes

4. Technologies Used:

- Java
- JUnit
- Maven
- Docker
- Kubernetes
- Jenkins
- GitHub

5. System Architecture:

Developer → GitHub → Jenkins → JUnit Testing → Build (Maven) → Docker Image → Kubernetes Deployment → Running Pods

6. Functional Requirements:

- Book appointment for a specific time slot
- Prevent double booking of the same slot
- Reject invalid or past time slots
- Allow valid appointment scheduling
- Automatically test booking logic

7. Non-Functional Requirements:

- Reliability: No duplicate bookings
- Scalability: Multiple users handled using Kubernetes
- Performance: Fast booking validation
- Maintainability: Easy updates via CI/CD

8. Appointment Scheduling Logic:

The system works based on:

- Each time slot can be booked only once
- If a slot is already booked → reject request
- Invalid time slots (past or incorrect format) are rejected
- Valid slots are successfully booked

9. Testing (JUnit)

Test Cases:

- Double booking prevention test
- Valid booking test
- Invalid time slot test

- Null input test

10. CI/CD Pipeline:

Steps:

1. Code is pushed to GitHub
2. Jenkins pulls the code
3. JUnit tests are executed
4. Application is built using Maven
5. Docker image is created
6. Application is deployed to Kubernetes

11. Docker Implementation:

Docker is used to containerize the system:

- Creates a portable environment
- Ensures consistent execution
- Simplifies deployment

Command:

```
docker build -t appointment-app .
```

12. Kubernetes Deployment:

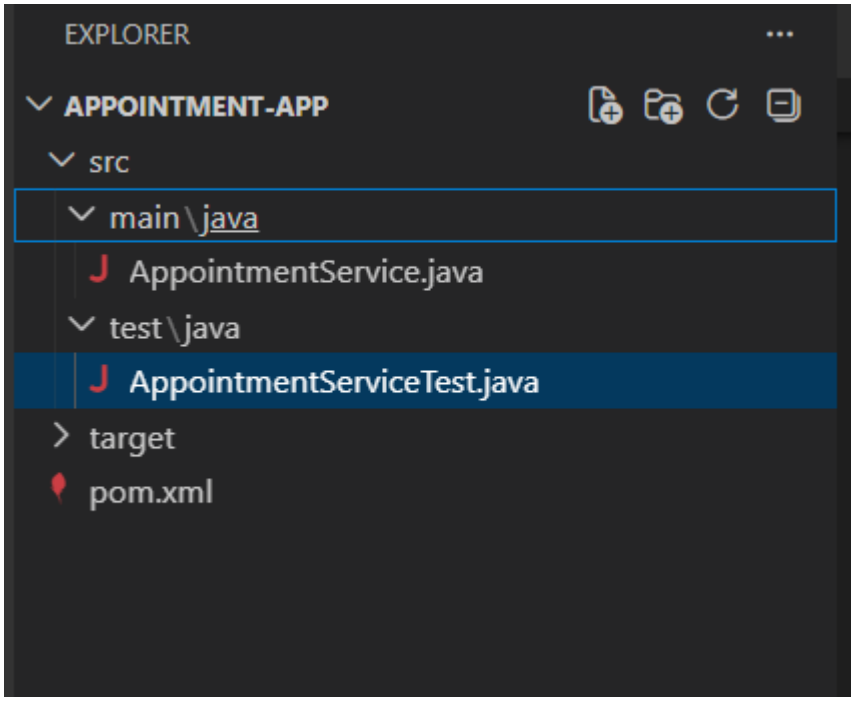
Kubernetes manages deployment:

- Uses deployment.yaml file
- Runs multiple replicas
- Ensures high availability

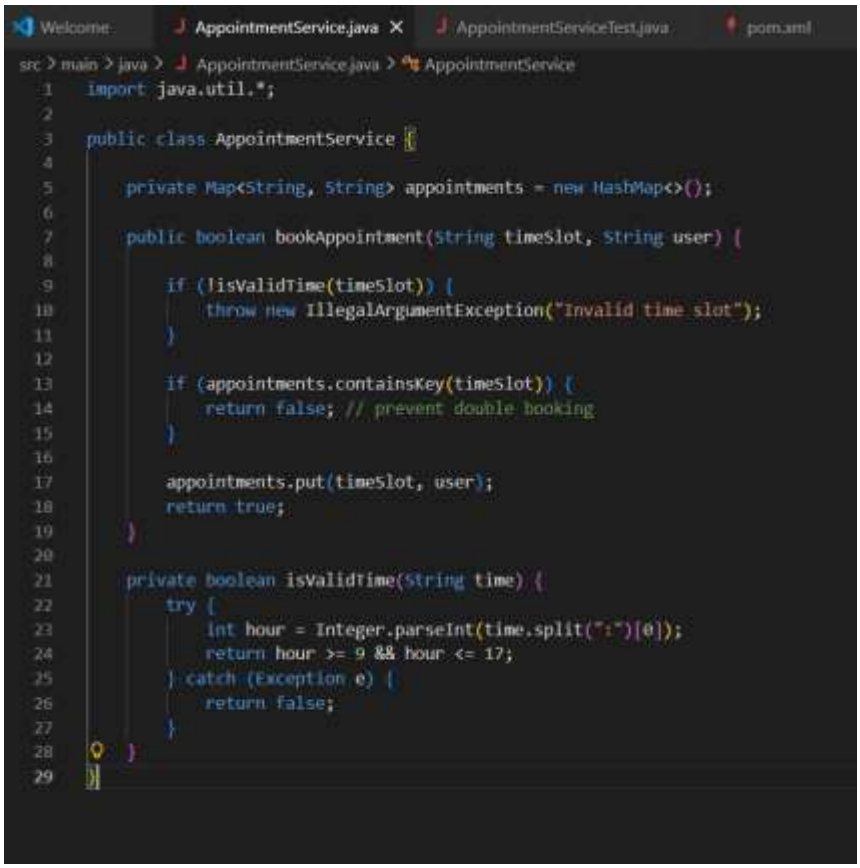
Commands:

```
kubectl apply -f deployment.yaml
kubectl get pods
```

Folder Structure:



AppointmentService.java:



AppointmentServiceTest.java:

AppointmentServiceTest.java:

Welcome | AppointmentService.java | AppointmentServiceTest.java | pom.xml

src > test > java > AppointmentServiceTest.java > AppointmentServiceTest

```java
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class AppointmentServiceTest {

    AppointmentService service = new AppointmentService();

    @Test
    void testDoubleBooking() {
        boolean first = service.bookAppointment(timeSlot: "10:00", user: "User1");
        boolean second = service.bookAppointment(timeSlot: "10:00", user: "User2");

        assertTrue(first);
        assertFalse(second);
    }

    @Test
    void testInvalidTimeSlot() {
        assertThrows(IllegalArgumentException.class, () -> {
            service.bookAppointment(timeSlot: "25:00", user: "User");
        });
    }
}
```

Pom.xml:

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.demo</groupId>
    <artifactId>appointment-app</artifactId>
    <version>1.0</version>
    <packaging>jar</packaging>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <junit.version>5.10.1</junit.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>${junit.version}</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
    <plugins>

        <!-- Compiler -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>
        </plugin>
```

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"

    <build>
    <plugins>

        <!-- Compiler -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>
        </plugin>

        <!-- JAR plugin (VERY IMPORTANT) -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.4.1</version>
            <configuration>
                <archive>
                    <manifest>
                        <mainClass>AppointmentService</mainClass>
                    </manifest>
                </archive>
            </configuration>
        </plugin>

    </plugins>
</build>

</project>
```

<u>Run Command:</u>

mvn clean test

```
PS C:\Users\sivas\downloads\appointment-app> mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:appointment-app >-----------------------
[INFO] Building appointment-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.2.5/maven-surefire-plugin-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.2.5/maven-surefire-plugin-3.2.5.pom (5.3 kB a
t 18 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.2.5/surefire-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.2.5/surefire-3.2.5.pom (22 kB at 439 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.pom (8.1 kB at
197 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom (7.5 kB at 170 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.jar (33 kB at 6
46 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/5.10.1/junit-jupiter-5.10.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/5.10.1/junit-jupiter-5.10.1.pom (1.2 kB at 70 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-api/5.10.1/junit-jupiter-api-5.10.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-api/5.10.1/junit-jupiter-api-5.10.1.pom (1.2 kB at 70 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/opentest4j/opentest4j/1.3.0/opentest4j-1.3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/opentest4j/opentest4j/1.3.0/opentest4j-1.3.0.pom (2.0 kB at 62 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-commons/1.10.1/junit-platform-commons-1.10.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-commons/1.10.1/junit-platform-commons-1.10.1.pom (2.0 kB at
67 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apiguardian/apiguardian-api/1.1.2/apiguardian-api-1.1.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apiguardian/apiguardian-api/1.1.2/apiguardian-api-1.1.2.pom (1.5 kB at 37 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-params/5.10.1/junit-jupiter-params-5.10.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-params/5.10.1/junit-jupiter-params-5.10.1.pom (3.0 kB at 73 kB
/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-engine/5.10.1/junit-jupiter-engine-5.10.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-engine/5.10.1/junit-jupiter-engine-5.10.1.pom (3.1 kB at 73 kB
/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-engine/1.10.1/junit-platform-engine-1.10.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-engine/1.10.1/junit-platform-engine-1.10.1.pom (3.2 kB at 89
 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter/5.10.1/junit-jupiter-5.10.1.jar
```

```
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-engine/1.10.1/junit-platform-engine-1.10.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-commons/1.10.1/junit-platform-commons-1.10.1.jar (106 kB at 4
799 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-params/5.10.1/junit-jupiter-params-5.10.1.jar (306 kB at 3.8 M
B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-api/5.10.1/junit-jupiter-api-5.10.1.jar (211 kB at 719 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/jupiter/junit-jupiter-engine/5.10.1/junit-jupiter-engine-5.10.1.jar (245 kB at 882 k
B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-engine/1.10.1/junit-platform-engine-1.10.1.jar (205 kB at 52
5 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ appointment-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ appointment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\Downloads\appointment-app\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ appointment-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ appointment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\Downloads\appointment-app\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ appointment-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ appointment-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/maven-surefire-common/3.2.5/maven-surefire-common-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/maven-surefire-common/3.2.5/maven-surefire-common-3.2.5.pom (6.2 kB
at 101 kB/s)
```

```
Windows PowerShell
at 101 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-api/3.2.5/surefire-api-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-api/3.2.5/surefire-api-3.2.5.pom (3.5 kB at 80 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-logger-api/3.2.5/surefire-logger-api-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-logger-api/3.2.5/surefire-logger-api-3.2.5.pom (3.1 kB at 9
8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-shared-utils/3.2.5/surefire-shared-utils-3.2.5.pom
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at 97 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-api/3.2.5/surefire-extensions-api-3.2.5.pom
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kB at 85 kB/s)
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Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-spi/3.2.5/surefire-extensions-spi-3.2.5.pom (3.8
kB at 97 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-util/1.0.0.v20140518/aether-util-1.0.0.v20140518.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-util/1.0.0.v20140518/aether-util-1.0.0.v20140518.pom (2.2 kB at 66 k
B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether/1.0.0.v20140518/aether-1.0.0.v20140518.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether/1.0.0.v20140518/aether-1.0.0.v20140518.pom (10 kB at 369 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-api/1.0.0.v20140518/aether-api-1.0.0.v20140518.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aether-api/1.0.0.v20140518/aether-api-1.0.0.v20140518.pom (1.9 kB at 56 kB/
s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.1.1/maven-common-artifact-filters-3.1
.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.1.1/maven-common-artifact-filters-3.1.
1.pom (5.8 kB at 132 kB/s)
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Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-artifact/3.2.5/maven-artifact-3.2.5.pom (2.1 kB at 60 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven/3.2.5/maven-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven/3.2.5/maven-3.2.5.pom (22 kB at 497 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/25/maven-parent-25.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/25/maven-parent-25.pom (37 kB at 784 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/15/apache-15.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/15/apache-15.pom (15 kB at 311 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-core/3.2.5/maven-core-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-core/3.2.5/maven-core-3.2.5.pom (8.1 kB at 202 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-settings/3.2.5/maven-settings-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-settings/3.2.5/maven-settings-3.2.5.pom (2.2 kB at 39 kB/s)
```

```
1.981 kB/s)
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppointmentServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.042 s -- in AppointmentServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ appointment-app ---
[INFO] Building jar: C:\Users\sivas\Downloads\appointment-app\target\appointment-app-1.0.jar
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ appointment-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom (2.2 kB at
2.4 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom (23 kB at 340 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom (2.2 kB at 8
8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom (7.8 kB at 196 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar (196 kB at
653 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 783 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (137 kB at 8
```

```
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom (7.8 kB at 196 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar (196 kB at
 653 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 783 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (157 kB at 5
38 kB/s)
[INFO] Installing C:\Users\sivas\Downloads\appointment-app\pom.xml to C:\Users\sivas\.m2\repository\com\demo\appointment-app\1.0\appointment-app-1.0.pom
[INFO] Installing C:\Users\sivas\Downloads\appointment-app\target\appointment-app-1.0.jar to C:\Users\sivas\.m2\repository\com\demo\appointment-app\1.0\appo
intment-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.283 s
[INFO] Finished at: 2026-04-06T20:56:58+05:30
[INFO] ------------------------------------------------------------------------
```

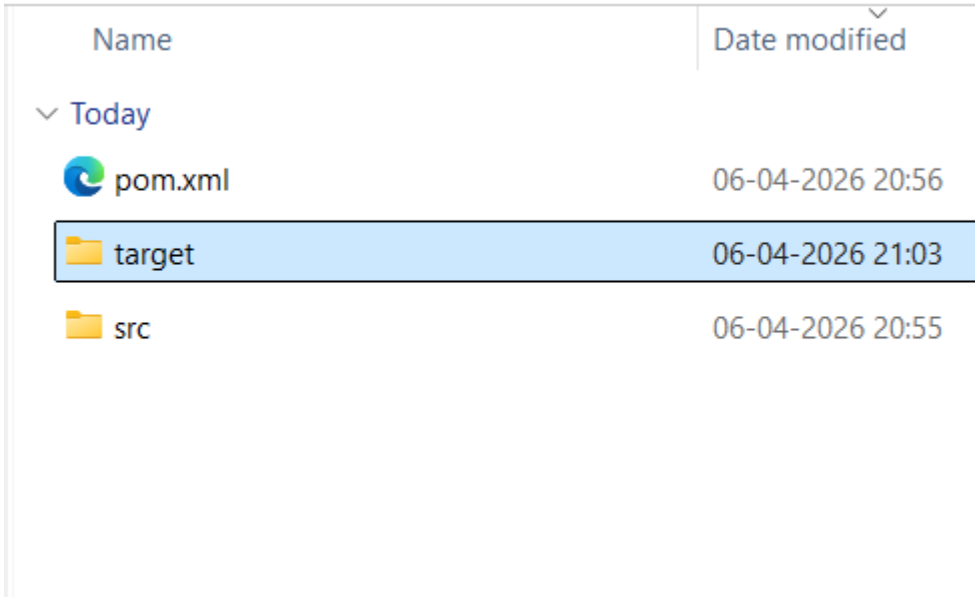
```
[INFO] ------------------------------------------------------------------------
PS C:\Users\sivas\downloads\appointment-app> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:appointment-app >-----------------------
[INFO] Building appointment-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ appointment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\Downloads\appointment-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ appointment-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ appointment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\Downloads\appointment-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ appointment-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ appointment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppointmentServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.044 s -- in AppointmentServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.890 s
```

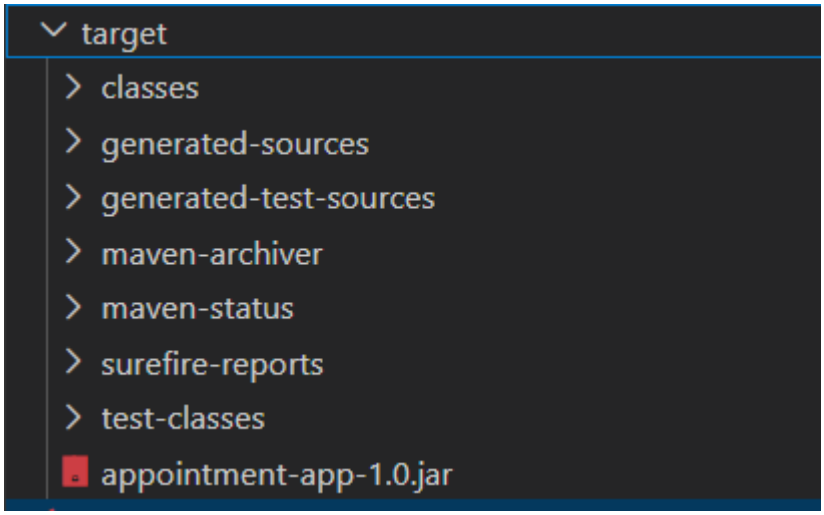
mvn clean package:

```
PS C:\Users\sivas\downloads\appointment-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:appointment-app >-----------------------
[INFO] Building appointment-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ appointment-app ---
[INFO] Deleting C:\Users\sivas\Downloads\appointment-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ appointment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\Downloads\appointment-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ appointment-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ appointment-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\Downloads\appointment-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ appointment-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ appointment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
```

```
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ appointment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppointmentServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.054 s -- in AppointmentServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ appointment-app ---
[INFO] Building jar: C:\Users\sivas\Downloads\appointment-app\target\appointment-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.067 s
[INFO] Finished at: 2026-04-06T21:03:54+05:30
[INFO] ------------------------------------------------------------------------
```

Target file:

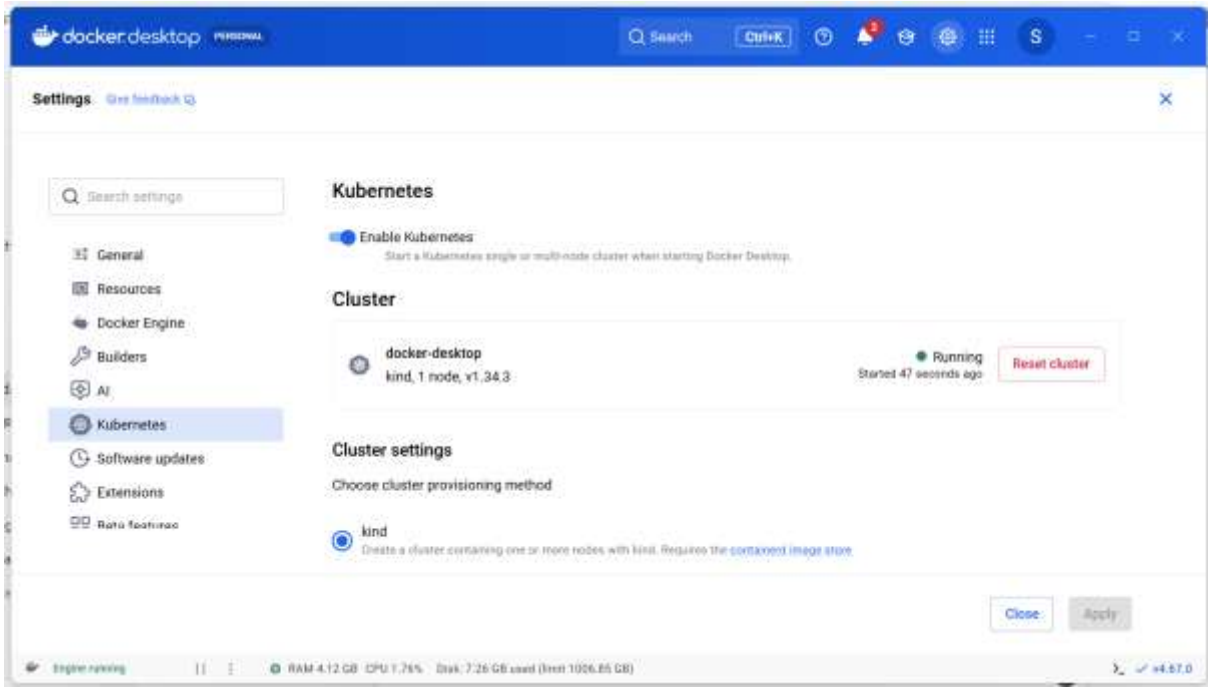




Dockerfile:

Build docker :

```
PS C:\Users\sivas\downloads\appointment-app> docker --version
Docker version 29.3.1, build c2be9cc
PS C:\Users\sivas\downloads\appointment-app> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\sivas\downloads\appointment-app> docker build -t appointment-app .
[+] Building 42.1s (9/9) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                0.1s
 => => transferring dockerfile: 167B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine                                                    0.7s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                              0.0s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/1] FROM docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e860ecd125a017594fd150d0942b998aa86957262f6dec4706512ba47bd4f8B3  30.5s
 => => resolve docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e860ecd125a017594fd150d0942b998aa86957262f6dec4706512ba47bd4f8B3  0.0s
 => => sha256:946c51efe0fd37b3c13616cebe0f5a7fcc6c82f60c8679d6cf5e8fddd8620fc9 2.28kB / 2.28kB                                      1.8s
 => => sha256:7cd2436f804e3c25b262ab1b7a9ff98f09c80c2b63359c7e6fc8b4c4364bb969 13MB / 13MB                                          2.3s
 => => sha256:82c5408f8c28f98209f7cb8b8fe7898ee7b430d17c840a361da6ef7827dba16f 144.76MB / 144.76MB                                  12.7s
 => => sha256:3a547da4fe1538a9de970334bfb2b6793ffde2bd214a847994365c731833eeff 21.32MB / 21.32MB                                    14.7s
 => => sha256:509802ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa8285cc21ac153 3.86MB / 3.86MB                                      1.5s
 => => extracting sha256:509802ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa8285cc21ac153                                           0.2s
 => => extracting sha256:3a547da4fe1538a9de970334bfb2b6793ffde2bd214a847994365c731833eeff                                           0.5s
 => => extracting sha256:82c5408f8c28f98209f7cb8b8fe7898ee7b430d17c840a361da6ef7827dba16f                                           1.6s
 => => extracting sha256:7cd2436f804e3c25b262ab1b7a9ff98f09c80c2b63359c7e6fc8b4c4364bb969                                           0.0s
 => => extracting sha256:946c51efe0fd37b3c13616cebe0f5a7fcc6c82f60c8679d6cf5e8fddd8620fc9                                           0.0s
 => [internal] load build context                                                                                                   0.1s
 => => transferring context: 2.56kB                                                                                                 0.0s
 => [2/3] WORKDIR /app                                                                                                              0.2s
 => [3/3] COPY target/*.jar app.jar                                                                                                 0.0s
 => exporting to image                                                                                                              0.2s
 => => exporting layers                                                                                                             0.1s
 => => exporting manifest sha256:23d54afd617c41c8aad97834laf891ad8ab957883dd88a65c58da27bec604986d                                  0.0s
 => => exporting config sha256:bf1e1e6863aa0429459351fd8ae266c5a57ed1c80ef3e859f8c12391e004e4                                       0.0s
 => => exporting attestation manifest sha256:f18d8b971a25e6ef67f4f3ec6a2611572d284e109dc75b828c6f17b240b5c006b                       0.0s
 => => exporting manifest list sha256:070cd4c5ab0ef15f0539631883fsebca8a783b207b0ab9000f2006cc96dbdd30                               0.0s
 => => naming to docker.io/library/appointment-app:latest                                                                           0.0s
 => => unpacking to docker.io/library/appointment-app:latest                                                                        0.0s
PS C:\Users\sivas\downloads\appointment-app>
```



Deployment.yaml:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: appointment-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: appointment
  template:
    metadata:
      labels:
        app: appointment
    spec:
      containers:
      - name: appointment
        image: appointment-app
        imagePullPolicy: Never
        ports:
        - containerPort: 8080
```

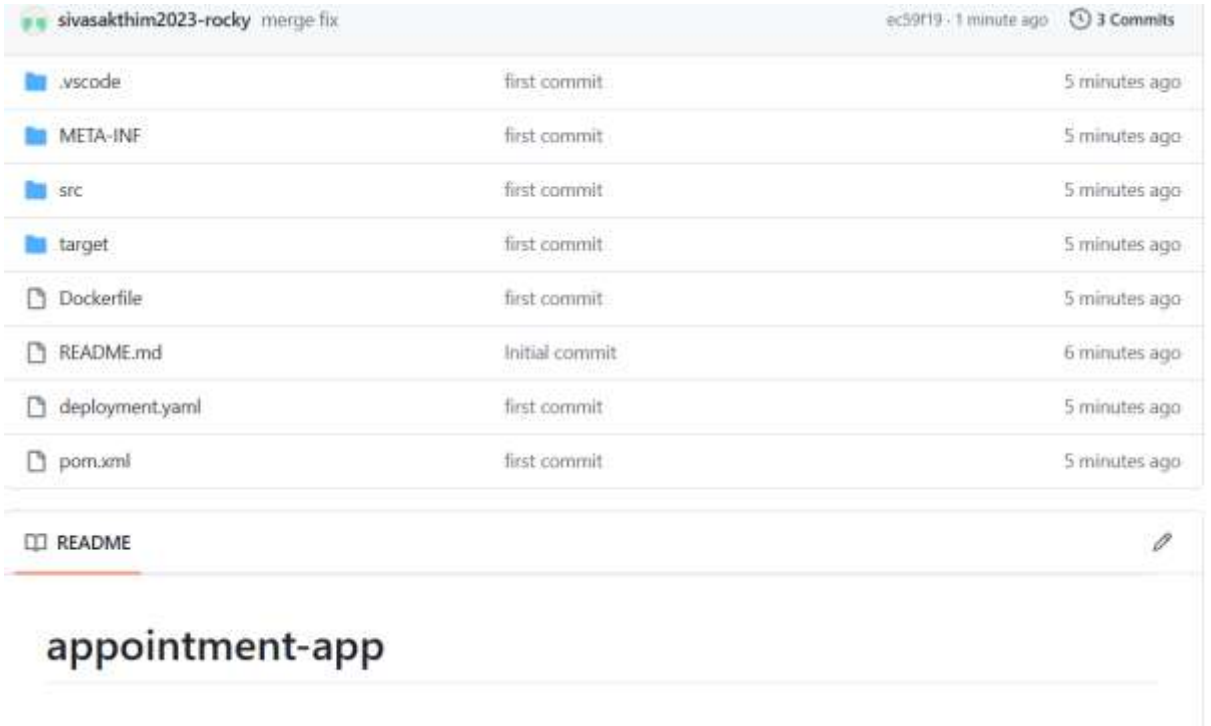
Run Commands:

```
PS C:\Users\sivas\downloads\appointment-app> docker run -p 9090:8080 appointment-app
Appointment App Running...
true
false
PS C:\Users\sivas\downloads\appointment-app> kubectl apply -f deployment.yaml
deployment.apps/appointment-app created
```

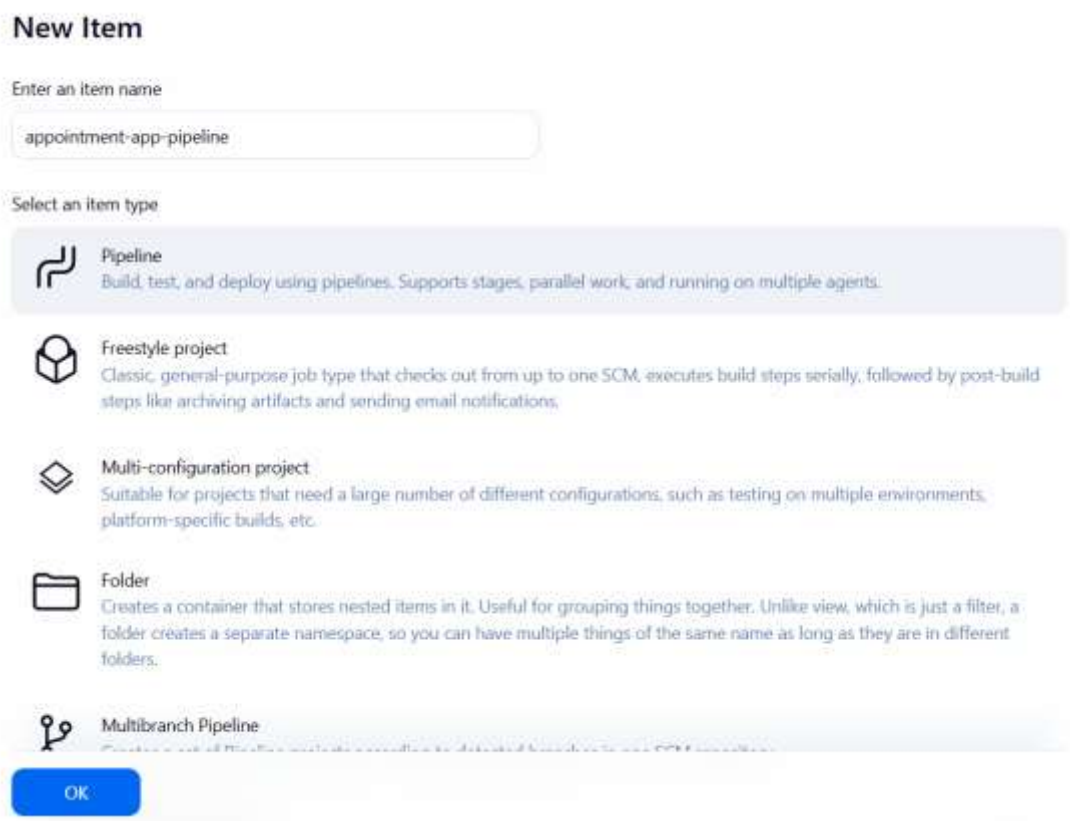
```
PS C:\Users\sivas\downloads\appointment-app> kubectl apply -f deployment.yaml
deployment.apps/appointment-app unchanged
PS C:\Users\sivas\downloads\appointment-app> kubectl get pods
NAME                                  READY   STATUS    RESTARTS        AGE
appointment-app-5795fb8c9b-28t7w      1/1     Running   5 (102s ago)    3m14s
appointment-app-5795fb8c9b-6cbgr      1/1     Running   5 (98s ago)     3m14s
PS C:\Users\sivas\downloads\appointment-app>
```

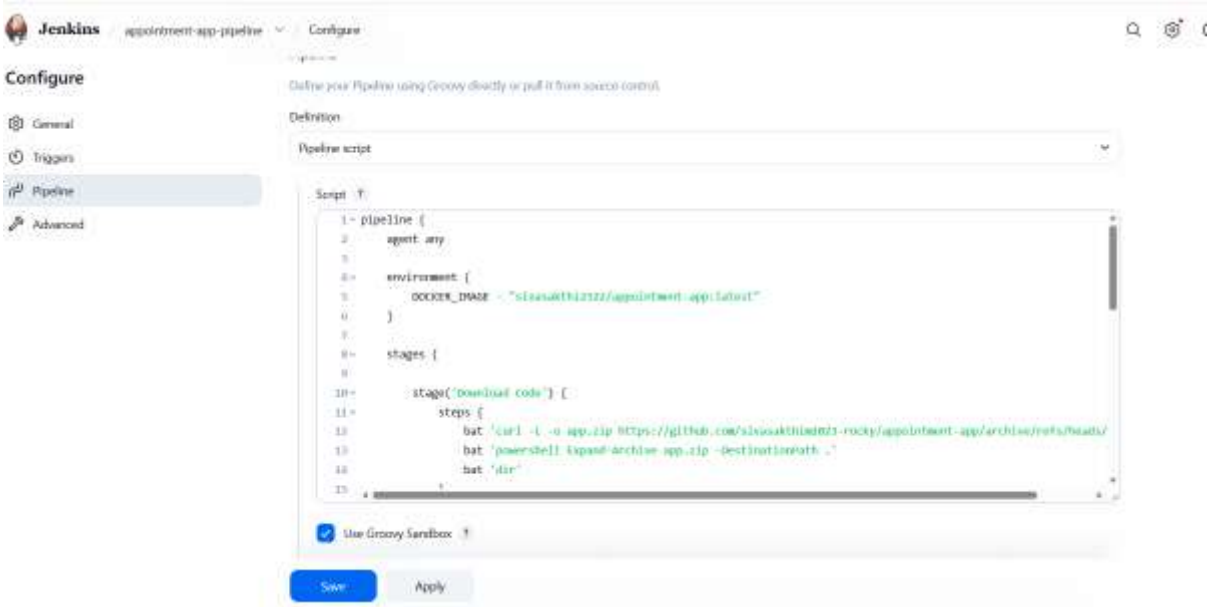
Push to github:

```
PS C:\Users\sivas\downloads\appointment-app> git add .
PS C:\Users\sivas\downloads\appointment-app> git commit -m "merge fix"
[main ec59f19] merge fix
PS C:\Users\sivas\downloads\appointment-app> git push origin main
Enumerating objects: 40, done.
Counting objects: 100% (40/40), done.
Delta compression using up to 12 threads
Compressing objects: 100% (29/29), done.
Writing objects: 100% (39/39), 9.63 KiB | 657.00 KiB/s, done.
Total 39 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/sivasakthim2023-rocky/appointment-app.git
   76c90ab..ec59f19  main -> main
PS C:\Users\sivas\downloads\appointment-app>
```



CI/CD Pipeline:





```
Started by user Sivasakthi
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Download Code)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline>curl -L -o app.zip https://github.com/sivasakthim2023-rocky/appointment-
app/archive/refs/heads/main.zip
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed

  0     0    0     0    0     0      0      0                              0
  0     0    0     0    0     0      0      0                              0
  0     0    0     0    0     0      0      0                              0

  0     0    0     0    0     0      0      0                              0
100  15349   0  15349   0     0   8452      0          00:01               0
100  15349   0  15349   0     0   8445      0          00:01               0
100  15349   0  15349   0     0   8439      0          00:01               0
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline>dir
 Volume in drive C is Windows
 Volume Serial Number is EA37-06AE

 Directory of C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline

07-04-2026  06:39    <DIR>          .
06-04-2026  22:45    <DIR>          ..
07-04-2026  06:39            15,349 app.zip
07-04-2026  06:39    <DIR>          appointment-app-main
               1 File(s)         15,349 bytes
               3 Dir(s)  11,580,276,736 bytes free
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline>cd appointment-app-main   && mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.demo:appointment-app >----------------------
[INFO] Building appointment-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ appointment-app ---
[INFO] Deleting C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline\appointment-app-main\target
```

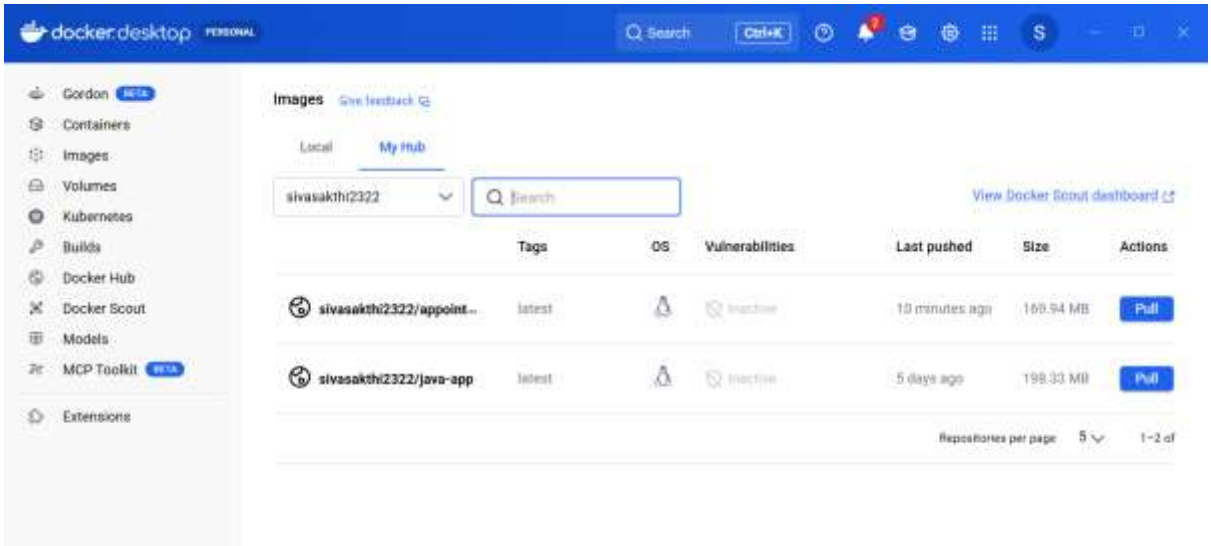
```
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ appointment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppointmentServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.070 s -- in AppointmentServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.567 s
[INFO] Finished at: 2026-04-07T06:39:38+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat
```

```
main\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ appointment-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ appointment-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppointmentServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.078 s -- in AppointmentServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ appointment-app ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\appointment-app-pipeline\appointment-app-main\target\appointment-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.537 s
[INFO] Finished at: 2026-04-07T06:39:44+05:30
[INFO] ------------------------------------------------------------------------
```

```
f8dc70e3afd0: Waiting
589002ba0eae: Waiting
82b06d8fada2: Waiting
3c5d41b4ad31: Waiting
3c5d41b4ad31: Waiting
f8dc70e3afd0: Waiting
589002ba0eae: Waiting
82b06d8fada2: Waiting
f8dc70e3afd0: Layer already exists
589002ba0eae: Layer already exists
3c5d41b4ad31: Waiting
3c5d41b4ad31: Waiting
82b06d8fada2: Pushed
3c5d41b4ad31: Pushed
latest: digest: sha256:8e09a91c3071e6b9e8e4771944766663330ef73a75676b7f5742a40b99042e6a size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

```
PS C:\Users\sivas\downloads\appointment-app> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
appointment-app-5795fb8c9b-gts78       1/1     Running   0          26m
appointment-app-5795fb8c9b-kl4nl       1/1     Running   0          26m
```

Docker Images:



Kubernetes deployment:

```
PS C:\Users\sivas\downloads\appointment-app> kubectl get svc
NAME                  TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
appointment-service   NodePort    10.96.157.150   <none>        80:30007/TCP   9s
kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        2m15s
```

13. Output:

- Appointment booking logic works correctly
- Double booking is prevented
- Invalid slots are rejected
- CI/CD pipeline executed successfully
- Application deployed in Kubernetes

14. Advantages

- Prevents scheduling conflicts
- Automated testing ensures reliability
- Faster deployment using CI/CD
- Scalable using Kubernetes

5. Limitations

- No graphical user interface
- Limited to basic scheduling logic
- No database integration

16. Future Enhancements

- Add user interface for booking
- Integrate database for storing appointments
- Add notifications (SMS/Email)
- Support multiple doctors/rooms

17. Real-World Relevance

This system is highly useful in:

- Clinic appointment systems
- Hospital scheduling
- Salon booking systems
- Online consultation platforms

18. Conclusion

The Appointment Booking System ensures accurate scheduling by preventing double bookings and validating time slots. Integration with CI/CD pipeline automates testing and deployment, making the system reliable, efficient, and scalable.